

HAT-P-54b

R-1383

Benchmark run · workflow W8-benchmark-deep · record deep-bench_HATP-54_201211 · created 2026-08-01T02:08:04+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2459194.7882 +/- 0.0024 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.198 +/- 0.055
Transit depth (Rp/Rs)^2	3.94 +/- 2.19 [%]
Orbital Inclination (inc)	83.6 +/- 2.26
Airmass coefficient 1 (a1)	0.887 +/- 0.07
Airmass coefficient 2 (a2)	0.12 +/- 0.15
Scatter in the residuals of the lightcurve fit is	7.86 %
Best Comparison Star	#1 - [369, 131]
Optimal Aperture	2.28
Optimal Annulus	14.42
Transit Duration (day)	0.038 +/- 0.021

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.198 ± 0.055 vs 0.1572 (deviation 0.53σ)
Duration fit vs published	0.038 d vs 0.075 d (deviation 1.27σ)
Mid-time O–C	-6.7 min
Depth SNR	2.6
Residual scatter	7.86 %

Light curve

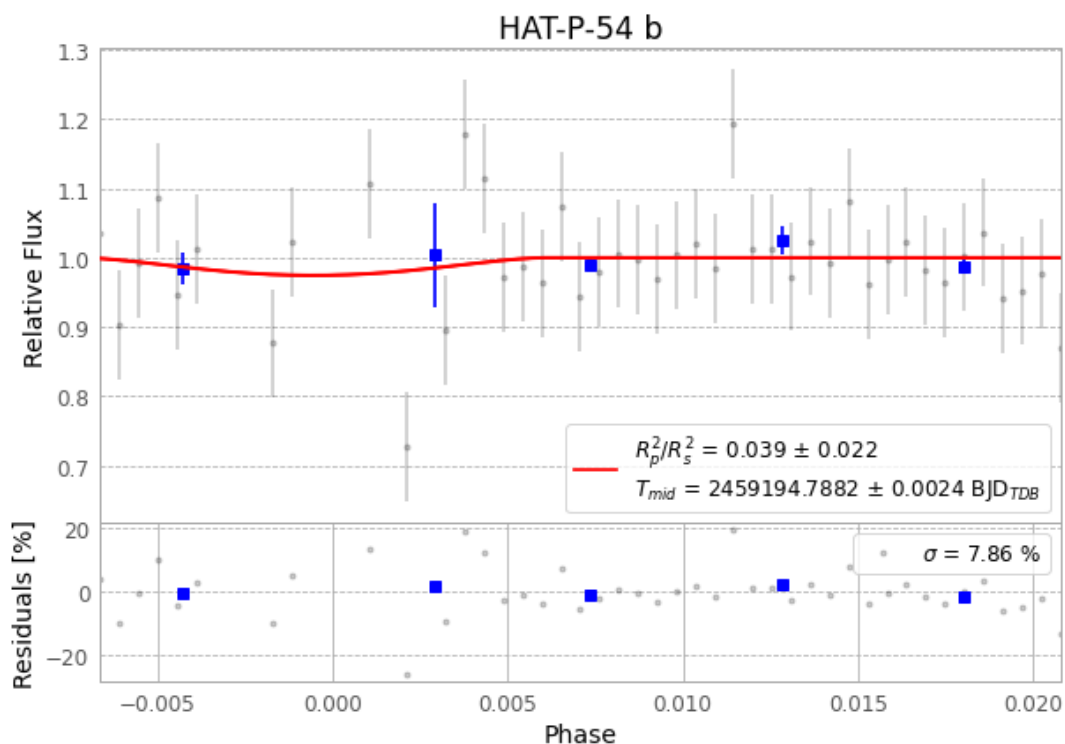


Figure 1 — FinalLightCurve_HAT-P-54 b_2020-12-10.png (SHA-256 f1e7693f28eb0160...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [478, 314], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 9066d8b35a8ba0b0...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

51 FITS frames (33 MB), HATP-54201211060914.FITS → HATP-54201211083914.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-54-201211.log (c1b02eaa981d...), FinalLightCurve_HAT-P-54 b_2020-12-10.png (f1e7693f28eb...), FinalLightCurve_HAT-P-54 b_2020-12-10.csv (5e23071ea5bc...)

Compute resources

Wall time 140.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-01 02:08Z.